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The Middle Jurassic Camaraca Formation, Arica, Chile: palaeomagnetism, K–Ar age dating and tectonic implications

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Summary. Palaeomagnetic and K–Ar whole rock age studies are reported from samples of the Camaraca Formation exposed near Arica (18.6° S, 70.3° W), Chile. The Camaraca Formation is composed of andesites and interbedded marine shales which yield a fauna placing the formation in the Bajocian–Callovian stages of the Jurassic. Our sampling sites span a stratigraphic thickness of 2 km and yield a reversal stratigraphy of: N–R–N. The K–Ar isochron method, applied to least altered samples from the formation, gives a sharply defined age of 157 Myr which is in agreement with the palaeontologically assigned age of the formation. Normal and reversed directions of remanent magnetization, isolated by af and thermal demagnetization methods, are statistically antiparallel. The pole position, computed from these directions of magnetization, is at 010° E, 70½° S (A_{95} = 6.0°). This pole position, when compared with the well-studied Chon Aike Formation of Argentina, suggests that the sampling region has undergone a 28° ± 28° counter-clockwise rotation about a local vertical axis. The large uncertainty (between 0° and 56°) in this estimate is due to the large scatter in the South American reference data. When compared with African Jurassic palaeopoles, with allowance made for the opening of the South Atlantic, a counter-clockwise rotation of 44° ± 14° is indicated.

Introduction

The pioneering work of Creer (1965) demonstrated that the pole with respect to South America in Mesozoic time was close to its present position. Thus, the Mesozoic magnetization in rocks from equatorial South America would have shallow inclination and be characterized dominantly by the horizontal component. Such conditions are optimal for applying the palaeomagnetic test for the orocline hypothesis (Carey 1955); an orocline being an orogenic system which has been flexed in plan to a horseshoe or elbow shape. The essential part of the oroclinal hypothesis is that the change in trend is an impressed strain

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(Carey 1955, 1976) rather than an original feature. One of the best examples claimed as an oroclinal bend is the change in trend of the Andean structures from near north–south to NW–SE at the Chile–Peru border. This orocline has been termed the Bolivian orocline by Carey (1958) and also referred to in his Fig. 27 as the Peruvian orocline. The initial objective of this study was a test of the oroclinal bending hypothesis using the palaeomagnetic technique as a tool.

Middle Jurassic andesites of the Camaraca Formation strike north–south in the extreme north of Chile, near Arica (Salas *et al.* 1966). In the immediately adjacent region of Peru the South American coastline changes abruptly from north trending to north-west-trending. Probable correlatives of the Camaraca Formation there are the Chocolate Volcanic Formation and the Guaneros Formation (Jaen & Ortiz 1963). The latter formations have a north-west regional tectonic strike. Accordingly oriented samples were collected from these formations to test whether the Bolivian orocline is indeed an impressed strain feature or whether the change in tectonic grain merely reflects a change in trend of the original depositional basin. Unfortunately, the rocks from the Peruvian exposures proved to be very unstable magnetically and no meaningful palaeomagnetic information could be extracted from them. On the other hand, the majority of samples from the Chilean Camaraca Formation proved to be moderately stable and we here report those results, together with the results of a K–Ar age study of some of them.

Geology

The rocks of the Arica Group consist of two formations: the primarily volcanic Camaraca Formation and the primarily sedimentary Los Tarros Formation (Salas *et al.* 1966). Ammonite assemblages place the Camaraca Formation in the Bajocian–Callovian stages of the Middle Jurassic whereas the Los Tarros Formation is assigned to the Oxfordian stage of the Upper Jurassic. The Arica Group is overlain discordantly by the Neocomian Atajaña Formation. All of our sampling sites are in the andesites of the Camaraca Formation. The thickness of the Camaraca Formation has been estimated at 1900 m (Salas *et al.* 1966) and 2800 m (Tobar, Salas & Kast 1968). Much of the uncertainty in the thickness of the formation stems from the fact that it is severely broken by numerous north-trending faults and also intruded by dykes neither of which are illustrated on Fig. 1. The stratigraphic relationship of the Camaraca Formation to older rocks is not known as the base of the formation is below sea-level. Older rocks represented by mica schists and gneisses of the Precambrian (?) Belen Schists are brought up in fault slices 100 km to the east of the main outcrop belt of the Arica Group (Salas *et al.* 1966). These schists are overlain unconformably by the Cretaceous–Tertiary Lupica Group rocks. However, because of the intervening cover of ignimbrite sheets of the Middle Tertiary Oxaya Formation, the relationship of the Lupica Group to the Arica Group is not known and thus the relationship of the Camaraca Formation to Andean basement is an open question. We dwell on this point because one of the results of this study is evidence that the Camaraca Formation has undergone an anticlockwise rotation in addition to the 20°–30° eastward tilting which is geologically observable.

Sampling sites (Fig. 1) are primarily in andesite flows of the Camaraca Formation. A few sites are in related diabase and microdiorite sheets and dykes. The flow rocks are grey, brownish grey and brownish red rocks commonly porphyritic in plagioclase and less commonly in pyroxene or the alteration products of pyroxene. Amygdular phases of flows are sparse. A few flows exhibit pillow structure. Mineralogically the flows consist mainly of plagioclase, clinopyroxene and oxides. Scarce biotite is in part altered to haematite.

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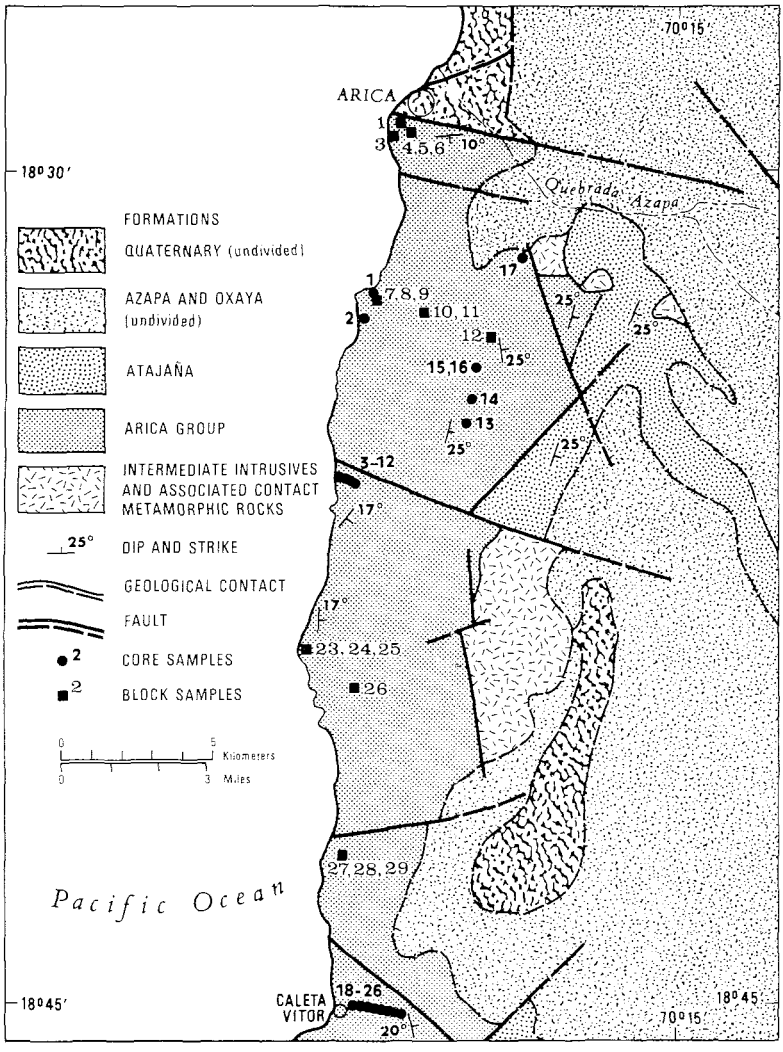


Figure 1. Geological map of a portion of the Department of Arica, Chile simplified from a portion of the map of Salas *et al.* (1966). Sample locations in the Camaraca Formation are indicated by squares (AR series) and solid dots (CA series). The location of the sampling region is shown as locality 9 on the map of South America in Fig. 7.

Olivine or the alteration products of olivine, serpentine and oxides, are sparsely present in a few flows. The intensity of secondary mineral alteration varies generally increasing from north to south. The least altered samples were taken immediately south of Arica, with chlorite, calcite and sparse sericitic alteration of the plagioclase being the only obvious secondary minerals. Southward across the east–west faults shown in Fig. 1, clinopyroxenes are partially replaced or wholly replaced by calcite, chlorite, epidote and possibly pumpellyite. Alteration of the plagioclases to the same assemblage of minerals together with white mica becomes more pronounced from north to south. Tremolite–actinolite as a replacement mineral appears in the south of the area (site 20, Fig. 1) in a dyke rock near Caleta Vitor. The mineralogical evidence is consistent with the contention of Salas *et al.* (1966) that most of the east–west faults cutting the Camaraca Formation are down-thrown to the north, the more southerly exposures thus representing deeper levels. Although lacking

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incipient metamorphic foliation, the more southerly exposures have undergone burial metamorphism to conditions approaching the greenschist facies.

The iron–titanium oxides, now greatly altered to maghaemite and haematite, were titanomagnetites in which the level of original high temperature deuteric oxidation was generally low. The majority of samples examined in reflected light with oil immersion exhibit relict titanomagnetite dominated by oxidation classes 1 to 3 as defined by Wilson & Haggerty (1966), Ade-Hall *et al.* (1968) and Haggerty (1976). In the extreme northern part of the sampling area, 'exsolved' ilmenite is still present, whereas in the southern part of the sampling area the included ilmenite lamellae are totally replaced by black sphene, a characteristic burial metamorphism feature typical of conditions from the base of the laumontite zone to the epidote zone (Ade-Hall, Palmer & Hubbard 1971). Thus the iron–titanium oxide petrography is consistent with that found in transmission microscopy and supports the view that the southern exposures of the formation have suffered more intense burial metamorphism.

Sampling and laboratory methods

An early collection of block samples (AR series) was made in 1967 as part of a reconnaissance sampling of various Mesozoic units of northern Chile. Preliminary analysis of core samples drilled from these 18 block samples yielded an encouraging result with both polarities being present and reasonably good grouping of the sample directions within each of these two polarity groups. A second collection (CA series) was made in 1974 using portable rock drills in which generally four independently oriented cores were obtained at each of 26 sites. In the second collection, most of the sampling was concentrated in two profiles (sites 3–12 and 18–26 of Fig. 1) where uninterrupted stratigraphic successions could be obtained. This sampling scheme attempted to gain some knowledge of the reversal frequency in the part of the Middle Jurassic represented by the Camaraca Formation. The azimuthal component of orientation of the field-drilled cores was determined with a sun compass and checked with a Brunton compass corrected for the local published magnetic declination. Agreement of the sun and magnetic bearings to within 3° indicates that there is no orientation problem with the older collection of block samples (AR series) which were oriented by magnetic methods only.

In the laboratory two to six specimens were drilled from each block sample and one or two specimens could be obtained from each field-drilled core. One polished thin section from each of the block samples and from one of the cores from each of the 26 field-drilled sites was prepared. In addition, small chips each weighing 0.5–1 g were machined for use in a Curie balance.

The early stages of the palaeomagnetic measurements were performed using a PAR spinner magnetometer. All measurements of remanent magnetization intensity and direction subsequent to the 30 mT af demagnetization step, and all measurements subsequent to thermal demagnetization steps, were performed with a Schonstedt DSM-1 spinner magnetometer. Alternating field demagnetization was carried out to a maximum of 110 mT (peak) using a three-axis tumbler and coil apparatus similar to that described by Roy, Reynolds & Sander (1973). Two to four specimens from each block sample and one specimen from each of the 102 field-drilled core samples were progressively demagnetized in steps to a routine maximum of 80 mT. The most representative value of the mean direction for a site or a sample was made by choosing the data set from the demagnetization level for which the specimen directions were most tightly clustered provided that the field exceeded 10 mT. In a few cases, a less well clustered level was chosen where visual inspection of the data

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indicated that an end-point had not quite been reached during the best grouped interval. Step-wise thermal demagnetization was carried out on selected samples and sites. In some experiments the thermal demagnetization was performed on specimens which had previously been subjected to the partial alternating field demagnetization treatment; in other experiments the thermal demagnetization was done on specimens which had not been so treated. The thermal demagnetization experiments were done in furnaces employing nested mumetal shielding. One of these demagnetizers is a Schonstedt TSD-1 model; the other is a hand-crafted model in which the specimens are slowly rotated along one axis during cooling. This rotation minimizes the possible acquisition of small PTRM's in one plane due to incomplete dc field cancellation. Temperature calibration experiments, temperature gradient measurements and duplicate runs in this furnace and in the TSD-1 indicate that temperatures (given on the thermal demagnetization plots) are conservatively estimated to be accurate to 5°C. J_{sat} versus T curves were obtained by heating and cooling small chips in air in a constant gradient field by a null method in a horizontal translation balance; at the null position the applied field used was 0.8 T.

Palaeomagnetic results

Approximately 35 per cent of the samples in the collection have median destructive fields (mdf) of 5 mT or less and only about 15 per cent of the collection have mdf's exceeding 40 mT. On the whole the collection was not an easy one to work with and the data from a number of sites had to be rejected (as noted in Table 1). This is particularly true in the field-drilled collection (CA series) which emphasized continuous section coverage which in some outcrops meant collecting from poorly exposed and weathered outcrops. The block samples (AR series) were taken from well exposed outcrops, all of which provided usable palaeomagnetic information.

Table 1. Palaeomagnetic directional data.

Site/ sample	N	af/T	R	k	D (°)	I (°)	D Structurally (°)	I corrected (°)	Pole position Long.	Lat.
AR-1	2	30 mT	1.998	504.0	009	−49½	358	−46½	62° W	81° S
AR-3	4	30 mT	3.998	1300.0	340½	−27	337	−20	36½° E	66° S
AR-4	2	30 mT	1.999	2093.0	346½	−25	343½	−19	45½° E	72° S
AR-5	2	10 mT	1.997	348.0	336½	−54	328½	−46½	4° W	60° S
AR-6	2	30 mT	1.999	955.0	004	−36½	352	−29	40° W	82° S
AR-7	2	625°C	1.963	27.0	150	+71½	147½	+51½	12° W	58° S
AR-7	2	15 mT	1.995	189.0	218½	+52	199	+43		
AR-8	2	40 mT	1.979	49.0	334	−45	332	−25	27° E	63° S
AR-9	3	80 mT	2.997	558.0	208	41½	195½	+30½	166° W	75° S
AR-9	3	625°C	2.995	448.0	206	+36½	196	+25		
AR-10	2	40 mT	1.999	3077.0	182	+43	173½	+40	16° W	83° S
AR-11	2	30 mT	1.996	258.0	204½	+01	204	+03	167° E	61° S
AR-11	2	unstable in thermal demagnetization								
AR-12	3	30 mT	2.997	821.0	315½	−63	308	−54	10½° W	41° S
AR-12	2	unstable in thermal demagnetization								
AR-23	3	30 mT	2.963	54.0	149½	+33½	141½	+24	23° E	53° S
AR-24	3	40 mT	2.879	17.0	183½	+60½	155½	+57		
AR-24	4	600°C	3.991	334.0	167	+44½	152½	+38½	8° E	64° S
AR-25	4	40 mT	3.991	339.0	166	+61	142	+53		
AR-25	4	600°C	3.958	71.0	168½	+50½	151	+44½	1° W	62° S

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Table 1. Palaeomagnetic directional data (*contd*).

Site/ sample	*N*	af/*T*	*R*	*k*	*D* (°)	*I* (°)	*D* Structurally corrected (°)	*I* (°)	Pole position Long.	Lat.
AR-26	4	50 mT	3.995	583.0	348½	−35	338	−30½	23° E	69° S
AR-26	2	400°C	1.998	524.0	343½	−29½	335½	−23½		
AR-27	2	70 mT	1.978	45.0	356	−25	348½	−22½	50° E	77° S
AR-28	3	25 mT	2.992	266.0	351	−53	331	−47½	6° W	62° S
AR-29	2	100 mT	1.993	136.0	148	+18	144½	+08½	38° E	53° S
CA-1	4	unstable in af demagnetization								
CA-2	4	10 mT	3.985	195.0	346½	−34	339	−26	30° E	69° S
CA-3	4	unstable in af and thermal demagnetization								
CA-4	4	unstable in af demagnetization								
CA-5	4	10 mT	3.919	37.0	349½	−45½	341½	−31½	22° E	72° S
CA-5	4	500°C	3.996	720.0	334	−43½	330	−28		
CA-6	4	unstable in af and thermal demagnetization								
CA-7	4	30 mT	3.997	1164.0	346	−54½	336	−39	7° E	67° S
CA-7	4	600°C	3.863	22.0	001	−68½	339½	−54½		
CA-8	4	15 mT	3.736	11.0	331½	−45½	327	−28½	20° E	58° S
CA-9	3	15 mT	2.993	288.0	349	−47	340½	−32	20° E	71° S
CA-10	3	30 mT	2.994	345.0	344	−46½	336½	−31	21° E	68° S
CA-11	4	80 mT	3.997	990.0	333½	−43½	329	−26½		
CA-11	4	500°C	3.996	720.0	334	−43½	329½	−27	23° E	60° S
CA-12	3	20 mT	2.985	130.0	329½	−48½	325	−31½	17° E	57° S
CA-13	4	30 mT	3.980	152.0	172½	+33	162	+19½	44° E	71° S
CA-13	4	550°C	3.962	79.0	180	+34	167½	+23		
CA-14	4	unstable in af demagnetization								
CA-15	4	20 mT	3.996	799.0	346½	−52½	321	−46½	2° W	53° S
CA-16	4	unstable in af demagnetization								
CA-17	4	20 mT	3.891	28.0	351½	−52½	325	−48	5° W	57° S
CA-18	4	unstable in af demagnetization								
CA-19	4	60 mT	3.911	34.0	034	−39	008	−41½	125° W	81° S
CA-20	4	unstable in af demagnetization								
CA-21	4	60 mT	3.951	61.0	062½	−46	025½	−60½		
CA-21	4	625°C	3.951	61.0	063	−47	024	−61½	105° W	59° S
CA-22	4	25 mT	3.630	8.0	072	−41	075½	−61	122° W	23° S
CA-23	4	unstable in af demagnetization								
CA-24	4	15 mT	3.955	66.0	358½	−46½	327½	−43	2° E	60° S
CA-25	4	unstable in af demagnetization								
CA-26	5	10 mT	4.953	85.0	005½	−49	341½	−52	23° W	68° S

Notes:

N is the number of *specimens* studied in the AR series and is the number of *core samples* studied in the CA series. The column labelled af/*T* indicates the cleaning treatment in either alternating fields in units of millitesla or by thermal demagnetization at the temperature indicated in degrees Celsius. *R* and *k* are the standard statistical parameters of Fisher (1953). *D* and *I* are the declination and inclination in degrees with north-seeking directions down taken as positive. VGP's, in the last two columns, are calculated from the structurally corrected declinations and inclinations.

The site/sample directional statistics are given in Table 1. For some samples or sites both af and thermally treated palaeomagnetic directions are given of which only one is used in the summary statistical information (Table 2; Fig. 2b and c). The af result or the thermal result was chosen, based on which was better grouped or which improved the overall formational mean grouping.

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Table 2. Summary of palaeomagnetic results.

	N	R	k	α_{95}	D (°)	I (°)	Pole Long.	Pole Lat.	Polar error dp (°)	Polar error dm (°)
Normal NRM *in situ*	24	22.125	12.3	8.8	354.2	−32.4				
Normal cleaned *in situ*	24	22.930	21.5	6.5	351.1	−46.3				
Normal cleaned tilt corrected	24	23.087	25.2	6.0	337.8	−37.4	9.6° E	69.0° S	4.3	7.1
Reverse NRM *in situ*	9	5.221	2.1	47.8	142.5	+71.5				
Reverse cleaned *in situ*	9	8.158	9.5	17.6	173.7	+39.5				
Reverse cleaned tilt corrected	9	8.188	9.9	17.3	164.2	+30.8	24.8° E	74.8° S	11.2	19.2
NRM *in situ* normal and reverse combined	33	26.207	4.7	13.0	351.8	−39.9				
Cleaned *in situ* normal and reverse combined	33	31.045	16.4	6.4	351.8	−44.5				
Cleaned tilt corrected, normal and reversed combined	33	31.208	17.9	6.1	339.6	−35.7	12.9° E	70.7° S	4.3	7.1
Mean of site poles	33	31.330	19.2				9.9° E	70.5° S	A_{95} = 5.9°	

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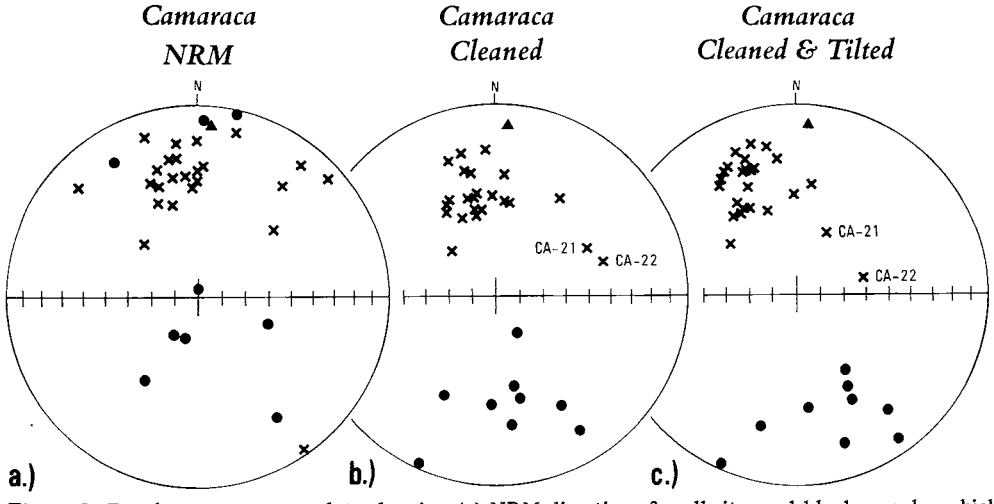


Figure 2. Equal area stereonet plots showing (a) NRM directions for all sites and block samples which proved to be stably magnetized by subsequent cleaning methods, (b) these same site/sample directions after magnetic or thermal cleaning, (c) after the application of the structural correction by rotating the directions in (b) by their field measured values of dip about the direction of strike. The triangle is the Earth's present field direction and X's (dots) represent north-seeking directions of magnetization on the upper (lower) hemisphere.

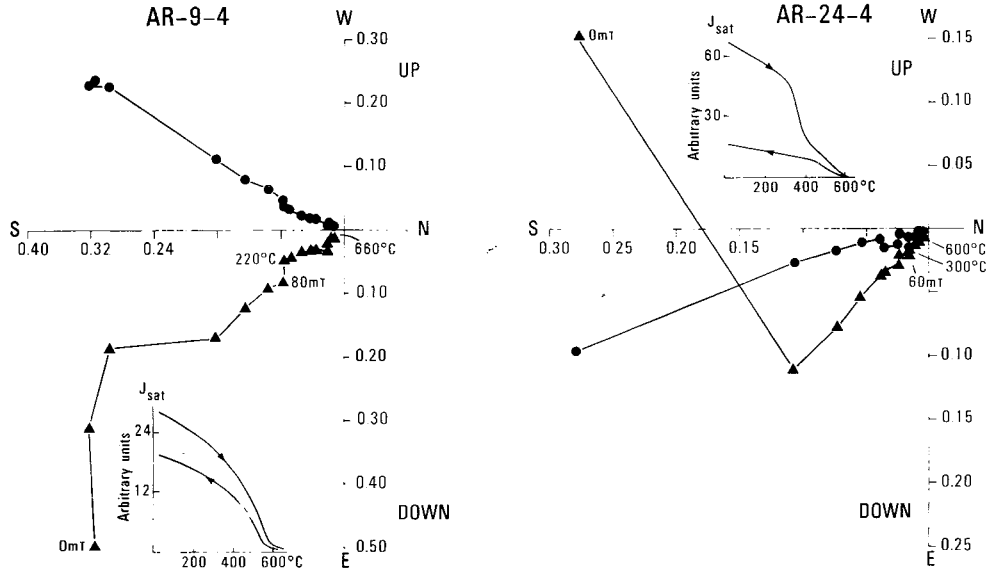


Figure 3. Orthogonal demagnetization projection (Zijderveld 1967) of the successive endpoints of the magnetization vector in two representative reversely magnetized specimens during progressive af demagnetization followed by thermal demagnetization. Structural tilt corrections have not been applied. Solid dots represent projections on the horizontal plane whereas triangles represent those on the north–south vertical plane. Units indicated for the axes are in A m^{-1} (10^{-3}emu cm^{-3}). Inset figures are J_{sat} versus T curves made from separate chips of these samples.

An inspection of Fig. 2(a) and (b) reveals that the treatment methods employed improve the grouping, and the sites can be classified as either normally or reversely magnetized. The application of the structural tilt correction (Fig. 2c) improves the clustering of each group but only marginally; the improvement is not statistically significant (Table 2). A significant

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improvement is not to be expected however as the dips of the flows, as measured from flow contacts and the attitude of interlayered sedimentary rock, are all shallow and homoclinal.

Excepting CA-21 and CA-22 (Fig. 2b and c) the normally magnetized flows are considerably better grouped than the reversely magnetized ones. CA-21, 22 and 23 are in stratigraphic order. CA-21 is very stably magnetized and is on the periphery of the normal group; CA-22 is a poorly grouped site and is excluded from the summary statistics of Table 2. CA-23 is a rejected site consisting of three unstable cores and one very stable reversely magnetized core. It seems probable that CA-21, 22 and 23 represent a poorly documented (normal to reverse) field transition.

Much of the detailed laboratory work was directed in searching for additional components of remanence which might possibly account for the greater dispersion of the reversed group. Examples from the results of these experiments are presented below. In Fig. 3 are shown two Zijderveld diagrams where the specimens have been treated by thermal demagnetization subsequent to partial af demagnetization. The directions move into the reversed group at very low af fields and then follow somewhat jagged paths to the origin. However, if one examines companion specimens from the same samples the nature of the deflections is not systematic. In Fig. 4 the effects of thermal demagnetization alone are compared with af plus thermal treatment for specimens from a reversely magnetized site. Specimens 46-2, 47-2 and 48-2 were examined by thermal demagnetization and it is noted that 300–400°C treatments serve to bring the specimen directions into concordance. It is inferred from the J_{sat} versus T curve, shown inset in Fig. 4, that over this temperature range maghaemite is demagnetized and/or converted to haematite. The directions remain reasonably well grouped to a temperature of 650°C after which scattering sets in without any systematic trend for the site as a whole. Repeat measurements of 49-1 subsequent to the 80 mT af step indicate that ARM's are troublesome. Subsequent thermal treatments fail to undo this effect and for the site as a whole the 550°C thermal result gives the best grouping (Table 1). The important point, however, is that using any higher temperature data set up to and including the 650°C run, would produce a very similar result.

Fig. 5 illustrates similar behaviour in a normally magnetized specimen from a normally magnetized site. For the site as a whole the 80 mT or the 500°C thermal treatment produces almost identical results. At temperatures above 500°C the mean site direction has lower precision but does not move significantly or systematically until 660°C, when final unblocking is obtained and the directions become random.

In summary, extensive laboratory analysis fails to detect unwanted secondary components of magnetization other than those which are eliminated at low af fields and at low blocking temperatures. Although the original high-temperature magnetic mineralogy has been modified by weathering the burial metamorphic processes, the initial directions of remanence can be recovered. Similar conclusions for similar rocks were reached by Marshall & Cox (1972). They showed that extensive mineralogical changes in weathered submarine basalt are accompanied by a number of changes in magnetic properties but, significantly, the primary remanence directions can still be recovered. A second argument that the directions obtained in the present study are indeed Middle Jurassic magnetic magnetizations is that the normal and reversed groups are statistically antiparallel (Table 2).

K–Ar dating

The experimental details of the K–Ar isochron method are summarized in Hayatsu & Carmichael (1977). Samples, selected from the collection on the basis of thin section examination, were crushed to pass 20 mesh. Powder finer than 100 mesh was discarded.

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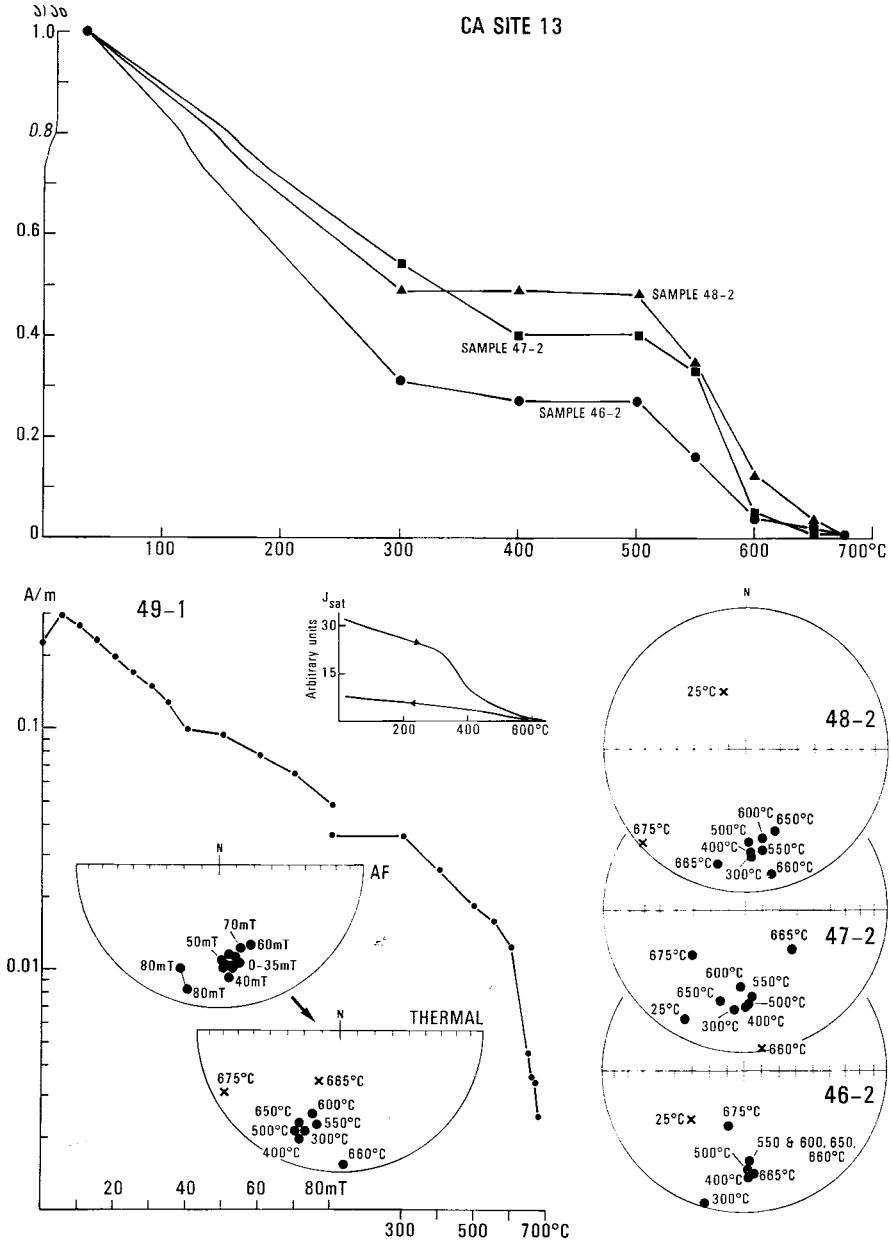


Figure 4. The effects of sequential af together with thermal demagnetization are compared with thermal demagnetization alone for four specimens from CA site 13. In specimens 46-2, 47-2 and 48-2 thermal demagnetization alone is used. In 49-1 thermal demagnetization follows af demagnetization. In the equal area plots X's represent directions on the upper hemisphere whereas dots are on the lower hemisphere. No structural correction applied.

Each sample was then split into two portions, one portion of which was further crushed to pass 100 mesh and used for potassium analysis. The other portion was loaded into an argon extraction system immediately after crushing, and baked for 18 hr at 190°C, followed by an argon extraction.

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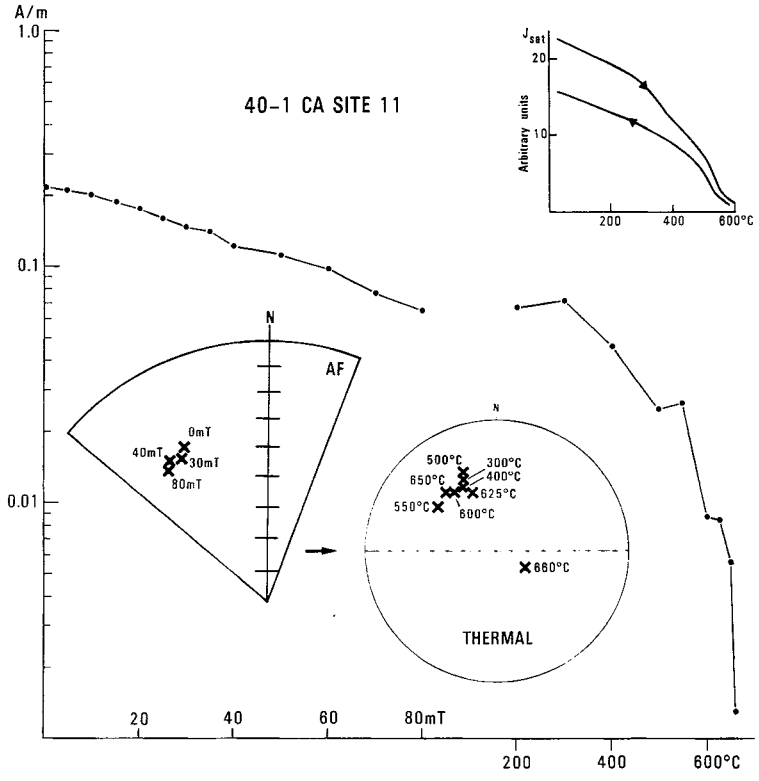


Figure 5. Progressive demagnetization of a normally magnetized specimen in which thermal demagnetization follows af demagnetization. All directions in the equal-area steronet plots are on the upper hemisphere and none are structurally corrected. Haematite is barely detectable in the J_{sat} versus T curve but magnetite (565°C) and maghaemite represented by the 350–400°C inflection in the heating curve are clearly distinguished.

Table 3. K–Ar analytical data.

Sample	K (per cent)	^{40}Ar rad. (10^{-5}cm^{3}g^{-1})	Age (Myr)	^{40}Ar total (10^{-5}cm^{3}g^{-1})	^{36}Ar (10^{-9}cm^{3}g^{-1})	^{40}Ar/^{36}Ar	^{40}K/^{36}Ar (10^5)
AR-1	0.994	0.418	105	0.719	10.19	706	0.65
AR-3	2.130	1.118	130	1.298	6.10	2126	2.34
AR-7	1.920	1.134	146	1.219	2.89	4220	4.45
AR-8	1.849	1.092	146	1.175	2.81	4185	4.40
AR-9	2.120	1.316	153	1.386	2.37	5840	5.99
AR-10	1.879	1.113	146	1.178	2.18	5401	5.77

The constants used were: $\lambda_e = 0.581 \times 10^{-10}$yr^{-1}; $\lambda_\beta = 4.962 \times 10^{-10}$yr^{-1}; atomic ^{40}K/K $= 1.167 \times 10^{-4}$.

The results are given in Table 3 and in Fig. 6. The ages calculated by the conventional method range from 105 to 153 Myr. Conventionally, the low ages would be attributed to argon loss. However, there is no correlation between the calculated age and the degree of alteration. Furthermore, on an isochron plot (Fig. 6), all points lie close to a 157 ± 4 Myr isochron with an initial ^{40}Ar/^{36}Ar ratio of 1 ± 105. Therefore the discordance in the ages calculated in the conventional way is most likely due to inadequate correction for non-radiogenic argon.

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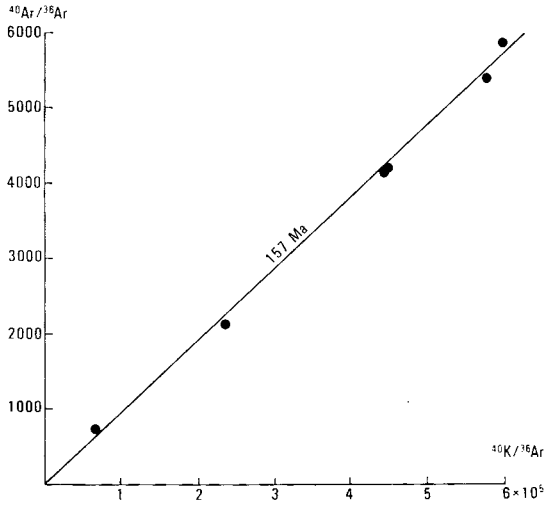


Figure 6. Isochron plot of the whole rock data indicates an age of 157 Myr for the Camaraca andesites.

The slope and the intercept of the isochron were calculated by the method developed by York (1969). The calculated initial ratio is close to the primordial $^{40}Ar/^{36}Ar$ ratio, suggesting that the argon was derived from the Earth's interior where the $^{40}K/^{36}Ar$ ratio would have to have been less than 700. Elements other than argon, however, may well have been derived from partial melting of, or contamination by, lithospheric plate material. Further study on the rare gas abundance pattern of these samples would seem fruitful.

Reversal stratigraphy and comparison with other South American poles

Both reversed and normally magnetized samples were used in constructing the K–Ar isochron of Fig. 6 and there is no resolvable difference in age between normal and reversed samples. Our field data suggest that the reversal stratigraphy from the exposed base to the highest stratigraphic samples is: N–R–N. We have no definitive evidence that more than one reversed zone is present within the Camaraca Formation. It is premature to make any attempt in correlating this reversed magnetozone with those that have been indicated from an analysis of global palaeomagnetic data (McElhinny & Burek 1971; Creer 1971; Pechersky & Khramov 1973) as these reversal schemes tend to be more generalized than is revealed by the more continuous record of reversal history given from the interpretation of marine magnetic anomalies (Larson & Pitman 1972). To date, linear magnetic anomalies have been assigned in time from the beginning of the Late Jurassic stopping just short of Camaraca Formation time. Should the stratigraphic limits of the formation be further refined by future palaeontological findings it may be possible, however, to integrate our magnetic data with Jurassic reversal history.

Well-controlled palaeomagnetic poles from rock units of undisputed Jurassic age are relatively scarce for South America (Table 4; Fig. 7a). The best studied unit is the Chon Aike Formation of Argentina (Valencio & Vilas 1970; Creer, Mitchell & Abou Deeb 1972; Vilas 1974). A Jurassic pole for Kimmeridgian sedimentary rocks, also from Argentina but based on untreated NRM data (Creer 1962) has large error limits (pole 3, Table 4). The older dyke suite of the Rio de los Molinos area, Argentina with an age of 140 Myr (Linares & Valencio 1975) lies near the Jurassic–Cretaceous boundary (pole 4, Table 4).

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Table 4. Jurassic palaeomagnetic poles for South America.

Pole no.	Rock unit	Locality	Age	Palaeomagnetic pole Lat.	Long.	dp (°)	dm (°)	A_{95} (°)	Reference
1	Chon Aike Formation	Camarones Bay, Argentina 44.8° S, 65.6° W	166 ± 5 Myr	82° S	129° W			12	Creer *et al.* (1972)
2	Chon Aike Formation	Puerto Deseado, Argentina 48° S, 66° W	161 Myr	84° S	056° E	6	7		Valencio & Vilas (1970)
3	Sediments*	Neuquen, Argentina 39° S, 71° W	Kimmeridgian	86° S	172° E			23	Creer (1962)
4	Dykes (older set)	Rio de los Molinos, Argentina 31.8° S, 64.5° W	140 Myr	79° S	008° E			18	Linares & Valencio (1975)
5	Chon Aike Formation	Estancia La Reconquista, Argentina 48.2° S, 68.3° W	Jurassic	69° S	177° W			13	Vilas (1974)
6	Sills in Las Cabras Formation	32° S, 69° W	204 Myr	74° S	094° W	11	18		Valencio (1969), Valencio & Vilas (1972)
7	Bolivar dyke set	Venezuela 7.5° N, 63.2° W	199 Myr	67° S	114° W	3	5		MacDonald & Opdyke (1974)
8	Guacamayas Group	Venezuela 8.9° N, 68.3° W	195 Myr	71° S	059° W	10	20		MacDonald & Opdyke (1974)
9	Camaraca Formation	Arica, Chile 18.6° S, 70.3° W	157 Myr	71° S	010° E			6	This study

* NRM results only.

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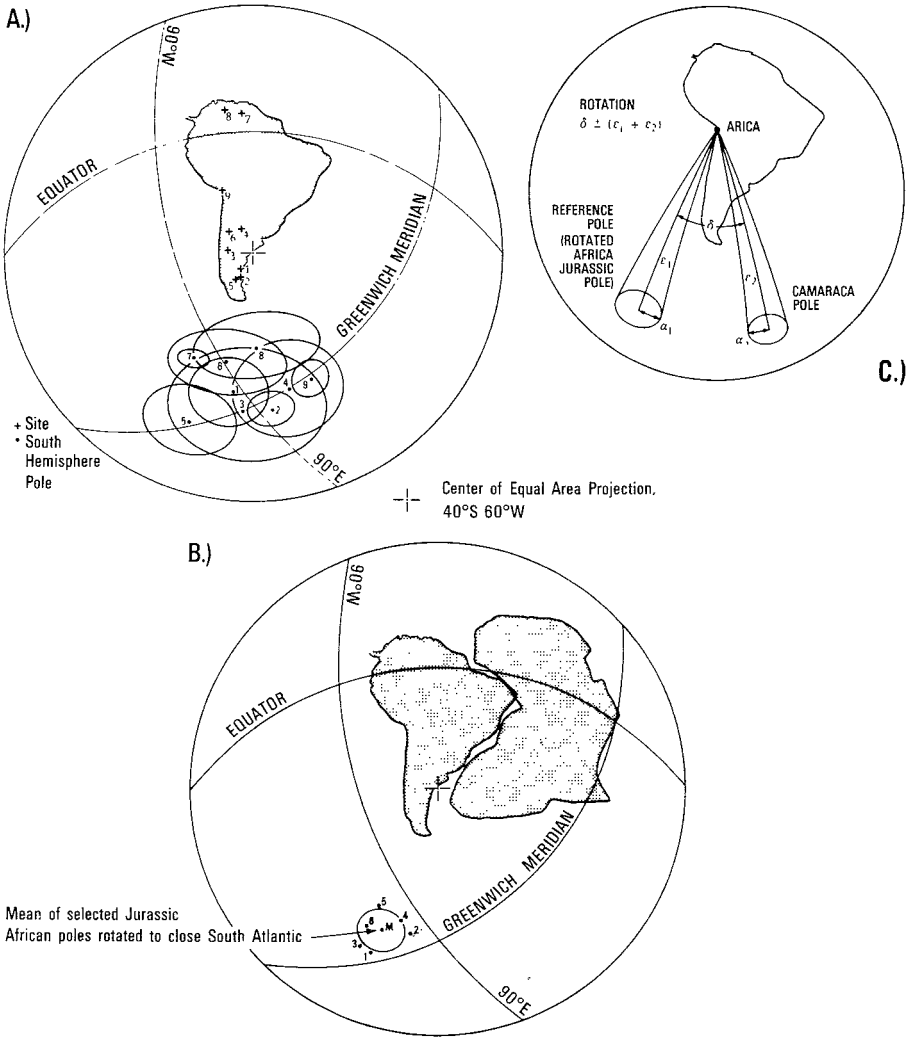


Figure 7. (a) Summary plot of Jurassic poles for South America. Older Jurassic poles (6, 7, 8) lie closer to the continent than younger Jurassic poles (1, 2, 3, 4, 5, 9). Both groups are spread along approximately arcuate trends. (b) Africa and its mean Jurassic pole (M) rotated about the Euler pole which best closes the South Atlantic (Bullard *et al.* 1965). Individual African poles are as follows: 1 Stromberg lavas (9.46, 9.47 and 9.48); 2 Mateke Hills Complexes (9.57); 3 Marangudzi Complex (9.59); 4 Hoachanas basalts (9.175); 5 Liberian diabase dykes and sills (9.172) and, 6 dolerite sills, Sahara (14.249). Pole reference numbers beginning with the number 9 are from Irving *et al.* (1976) and reference number (14.249) is from McElhinny & Cowley (1977). (c) The error in the rotation angle δ by which the sampling site must be rotated to bring the Camaraca pole into coincidence with the African reference pole is taken to be $\varepsilon_1 + \varepsilon_2$.

Much older poles, radiometrically dated and previously assigned to the Triassic, may also be Jurassic (poles 6, 7, 8, Table 4). The Triassic–Jurassic boundary, previously thought to be about 180 Myr (Kulp 1961; Harland, Smith & Wilcock 1964) has been recently assessed at about 210 Myr (Hedberg 1976; Armstrong 1978). Therefore, because of age uncertainty, the following units are also thought to be Early Jurassic: sills of the Las Cabras Formation, at 204 Myr, from Argentina (Valencio 1969; Valencio & Vilas 1972), the Bolivar dyke set,

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199 Myr, from Venezuela (MacDonald & Opdyke 1974) and the Guacamayas Group at 195 Myr, also from Venezuela (MacDonald & Opdyke 1974).

Spanning the Jurassic–Cretaceous boundary, but not included in the present compilation of Jurassic poles, is the Botucatu sandstone of Brazil (Creer, Embleton & Valencio 1970), and its stratigraphic equivalent, the Tacuarembo sandstone of Uruguay (Creer 1958). The reported Botucatu pole is based on NRM results and has large error limits (Creer *et al.* 1970). The Tacuarembo samples are of baked contacts affected by basalt flows of the younger Serra Geral Formation. The age of the magnetization therefore presumably represents the emplacement of the Serra Geral, and is post-Tacuarembo. The Botucatu and Tacuarembo might yet yield useful Jurassic data with further study.

Palaeomagnetic results from redbeds of possible Jurassic age from northern South America have been excluded from the present compilation. The Giron Formation of Colombia, for example, has been assigned to the Triassic (Creer *et al.* 1970) and to the Jurassic (Cediel, Ujueta & Caceres 1970). For want of more precise age data, both the Giron Formation and its possible correlative the La Quinta Formation of Venezuela are excluded for the time being.

Tectonic implications and conclusions

Jurassic poles of South America (Fig. 7a) reveal two trends: both older (poles 6, 7 and 8) and younger (poles 1, 2, 3, 4, 5, 9) Jurassic poles are scattered along approximately arcuate trends; and, older poles are closer to South America than are the younger poles. These results indicate: (1) significant tectonic rotation between sampling areas and (2) a northward shift of South America away from the south pole between Early and Late Jurassic times of about 10°.

The jurassic poles of South America were compared with those from Africa as compiled by Irving, Tanczyk & Hastie (1976) and McElhinny & Cowley (1977). The section of the African data was based on the following criteria: (1) that the stability of the magnetization was examined by one or more of af, thermal or chemical demagnetization techniques, (2) that the number of samples exceeded eight, (3) that the error about the calculated pole was less than 20°, (4) averages of previously reported poles were excluded, and (5) that localities from areas of Africa which have undergone post-Jurassic folding were excluded.

The poles so selected are plotted in Fig. 7(b). By rotating this African pole (83° E, 67° S, $A_{95} = 6.8°$) about the Euler pole which best closes the South Atlantic (Bullard, Everitt & Smith 1965) an estimate of the expected Jurassic pole for South America was obtained near 143° W, 68° S (M of Fig. 7b). Relative to this expected Jurassic pole for South America, the Camaraca strata have been rotated 44° ± 14° in a counter-clockwise sense. The error in this estimate is taken to be $\epsilon_1 + \epsilon_2$ of Fig. 7(c). The African reference pole is, however, based on data which have an average age about 20 Myr older than the rocks of the Camaraca Formation (Fitch & Miller 1970; Foland & Henderson 1976; Manton 1968; Gidskehaug, Creer & Mitchell 1975; Dalrymple, Gromme & White 1975). The discrepancy between the observed and expected poles might be attributed in part to age difference rather than local tectonic rotation.

The Camaraca Formation is close in age to the Chon Aike Formation of Argentina (Table 4). Although there is considerable scatter in the Chon Aike poles (poles 1, 2 and 5 of Table 4), a comparison of the mean of these (186° E, 82° S, $A_{95} = 21°$) with the Camaraca pole also indicates a counter-clockwise rotation of smaller amount but with considerable error (28° ± 28°). Thus it would appear that only part of the discrepancy between the rotated African reference pole and the Camaraca pole is possibly due to an age difference.

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The finding that the colatitude of Arica is not measurably different relative to the predicted and the observed younger poles of South America is consistent with the local rotation argument. This indirectly suggests that the rotation of the Arica section did not accompany micro-plate motion during which a former Arica micro-plate collided with South America. That is, the scatter observed, if due to tectonic effects, represents local tectonic movements rather than large scale plate tectonic movements. The limits of the rotated area are not known beyond the area sampled. The Arica rotation may be merely a local crustal or super-crustal plate-margin tectonic effect. Alternatively, it may represent movements of a larger crustal block, such as the Arequipa massif (Megard *et al.* 1971) on which Arica may rest. Further palaeomagnetic explorations in this section of the Andes could be most fruitful in resolving these large-scale problems.

Carey (1958) has noted the large tectonic deflection of the Andes at the latitude of Arica and named this the Bolivian orocline. Away from the oroclinal bend at the Chile–Peru border, the southern member of the Andes trends north–south. The northern member trends approximately north-westerly through Peru. Carey (1958) implies a 50° counter-clockwise rotation of the northern member relative to the southern member of the Bolivian orocline. Our results do not positively support this prediction as we have no data from the northern member. We find instead a 44° counter-clockwise rotation for the Arica area of the southern member of the orocline, relative to the Jurassic pole expected for South America. Clearly the tectonic rotations of the Andes are more complicated than the simple model of Carey would suggest.

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